

Zou Jiaju

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EDUCATION

• Beihang University (BUAA)

Sep 2024 – Jun 2027 (Expected)

M.Eng. in Mechanics | Research Institute of Aero-Engine (RIAE) |

Beijing, China

Fluid and Acoustic Engineering Laboratory (FAEL)

GPA: 3.74/4.00 (top 10%)

• Southern University of Science and Technology (SUSTech)

Sep 2020 – Jun 2024

B.Eng. in Aerospace Engineering | Department of Mechanics and Aerospace Engineering (MAE) |

Shenzhen, China

Advanced Aerodynamics and Aeroacoustics Laboratory (A3 Lab)

GPA: 3.76/4.00 (top 5%); Best Undergraduate Thesis Award & Outstanding Graduate Award

RESEARCH EXPERIENCE

• Inflow Calibration Monopole Source

BUAA | Apr. 2025 – Present

Advisors: Prof. Zhang Weiguang & Prof. Du Lin

- Developed a modular, broadband, omnidirectional sound source for aeroacoustic wind tunnel calibration and relevant acoustic measurements.
- Incorporated a face-to-face loudspeaker arrangement equipped with symmetric, low-drag smooth cabinets without externally exposed fasteners.
- Established electroacoustic COMSOL models to predict the far-field directivity of the annular sound radiator, with a maximum variability of ± 1 dB at 5 kHz within $\pm 60^\circ$ off-axis.
- Fabricated a prototype for a planned wind tunnel experiment and applied for a Chinese patent.

• Fan Inlet Noise Reduction via Intake Geometry Optimization

BUAA | Feb. 2025 – Present

Advisor: Prof. Du Lin

- Designed a converging-diverging segment within the fan intake to suppress the near cut-off noise.
- Performed simulations in COMSOL to predict transmission loss and evaluate acoustic performance.
- Validated the proposed noise control approach through measurements on a fan test rig, demonstrating tonal noise reduction of over 10 dB with negligible penalty in flow rate.
- Prepared a manuscript based on this work and submitted it to a peer-reviewed journal (in revision).

• Ground Noise Measurements

BUAA | Oct. 2024 – Present

Advisors: Prof. Du Lin & Prof. Zhang Weiguang

- Conducted indoor and outdoor experiments to quantify the effects of ground reflection and microphone installation on noise measurement.
- Determined correction factors to provide guidance for more accurate ground noise measurements.
- Presented the research findings at an international conference (Aeroacoustics 2026), including both full paper and oral presentation.

• Surface Pressure Fluctuation Measurements

SUSTech, A3 Lab | Mar. 2023 – Apr. 2024

Advisor: Prof. Liu Yu

- Arranged and calibrated a surface pressure measurement system using in situ pressure transducers.
- Captured surface pressure fluctuations inside a deep cavity and in the slit of a slitted airfoil.
- Contributed to peer-reviewed conference and journal publications in aeroacoustics.

• Acoustic Behavior of Corrugated Pipes with Mean Flow

SJTU, Shanghai | Jun. 2023 – Aug. 2023

Advisors: Prof. Wang Peng & Ph.D. Student Xu Zhaoyang

Undergraduate Summer Exchange Program

- Investigated plane wave propagation in corrugated pipes under co-flow and counter-flow conditions.
- Modeled the corrugated pipe as an effective medium and determined transmission and reflection coefficients using the two-microphone method both numerically and experimentally.

ENGINEERING PRACTICE

- **Low-Noise Leaf Blower Development**

Segway-Ninebot | Dec. 2025 – Apr. 2026

- Benchmarked commercial leaf blowers through psychoacoustic sound quality analysis.
- Designed and fabricated a leaf blower prototype as a test platform for blade design validation.

ACADEMIC PAPERS

Journal Articles

- **Zou, J.**, Yan, R., Du, L.* , Liu, G., Sun, X., & Feng, J. (2026). Near Cut-Off Noise Control Approach via a Converging-Diverging Segment. *Journal of Vibration and Acoustics*, in revision.
- Shang, H., **Zou, J.**, Du, L.* , Zhang, W., Wang, Z., & Sun, X. (2026). Unexpected High-Intensity Tone Noise from Wall-Mounted Finite Slitted Airfoil. *Aerospace Science and Technology*, 177(B), 112295 (JCR Q1). [Link](#)

Conference Papers

- **Zou, J.**, Du, L., Zhang, W.* , Wang, Z., Yang, M., & Wang, M. (2026). Experimental Study of Ground Reflection and Microphone Installation Effects on Noise Measurements. In *Proceedings of the 32nd AIAA/CEAS Aeroacoustics Conference*, AIAA Paper 2026-3551, Brussels, Belgium, 26–29 May 2026. [Link](#)
- Shang, H., **Zou, J.**, Du, L.* , Wang, Z., & Sun, X. (2026). Experimental Investigation of the Aeroacoustics of an Airfoil Undergoing Flow-Induced Vibration. In *Proceedings of the 32nd AIAA/CEAS Aeroacoustics Conference*, AIAA Paper 2026-3569, Brussels, Belgium, 26–29 May 2026. [Link](#)
- Naseer, M. R., Li, L., **Zou, J.**, Leung, R. C.* , Liu, Y.* , & Maryami, R. (2024). Experimental Study of Deep Cavity Noise Suppression by Surface Compliance. In *Proceedings of the 30th AIAA/CEAS Aeroacoustics Conference*, AIAA Paper 2024-3280, Rome, Italy, 4–7 June 2024. [Link](#)
- Li, L., Liu, G., **Zou, J.**, & Liu, Y.* (2024). Experimental Study on Fluid-Structure-Acoustic Interaction of a Cylinder with Flexible Splitter Plates. In *Proceedings of the 30th AIAA/CEAS Aeroacoustics Conference*, AIAA Paper 2024-3338, Rome, Italy, 4–7 June 2024. [Link](#)

PATENT APPLICATIONS

- Zhang, W., **Zou, J.**, Du, L., Fan, Z., Wang, Z., & Qiu, X. (2026). All-Directional Modularized Wind Tunnel Standard Sound Source. Chinese Patent CN121985249A. Patent pending.
- Du, L., **Zou, J.**, Wang, Z., Zhang, W., Wang, X., & Sun, X. (2025). Noise Reduction Tube. Chinese Patent CN120954369A. Patent pending.
- Wang, Z., **Zou, J.**, Du, L., Wang, D., Wang, M., & Yang, M. (2025). Adjustable-Angle High-Precision Spherical Sound-Field Generation and Positioning System. Chinese Patent CN120213200A. Patent pending.

SKILLS

- **Languages:** English (IELTS Academic 7.5), Chinese (Native)
- **Programming:** MATLAB, LabVIEW, Python, Java
- **Mechanical Design:** SolidWorks, AutoCAD, Siemens NX
- **Electronics:** EasyEDA, PCB design and soldering
- **Simulation:** COMSOL Multiphysics, ANSYS Workbench, OpenVSP
- **Scientific Illustration:** OriginLab, Adobe Creative Suite